

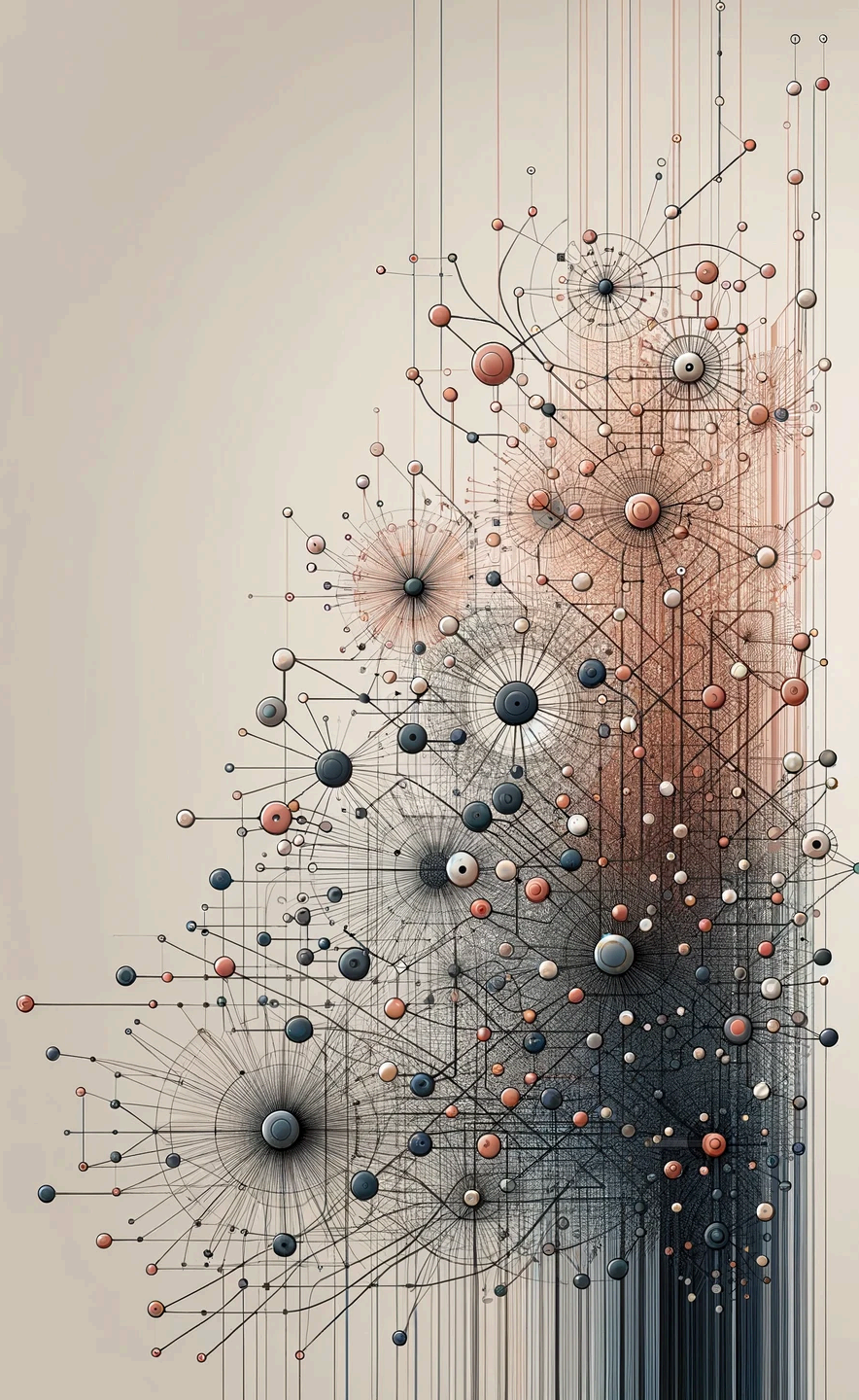


UNIVERSITÀ  
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# Analisi e Visualizzazione delle Reti Complesse

## NS22 - Compartmental Models in Computational Epidemiology

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## Outline

1. **Foundations** — what computational epidemiology is, what it is for, the modelling vocabulary
2. **The compartmental framework** — splitting a population into states; common assumptions
3. **First models** — SI (no recovery) and SIS (recovery without immunity)
4. **The SIR model** — recovery with immunity; epidemic threshold; key measures
5. **Extensions** — SEIR, SIRS, SEIRS, demography, age structure
6. **Bridge to networks** — what mean-field assumes, and what it misses

## A short story — what does an outbreak look like?

Think of any infectious disease — flu, measles, COVID-19. The story always has the same shape:

1. A few people are infected
2. They meet others; some of those others catch it
3. Cases grow — slowly at first, then **explosively**
4. Cases peak when there are too few susceptibles left to keep up
5. Cases fall — the outbreak ends

We want to **describe**, **predict**, and **intervene** in that story. Turning a curve into a system of equations is what compartmental models do.

## What we will be able to compute

By the end of this lecture, you will be able to answer — with formulas — the questions every public-health team faces:

- **How fast** does the outbreak grow at the start?
- **Will it die out** on its own, or persist forever?
- **How big** will the peak be — and **when** does it hit?
- **How many people** will be infected in total?
- **What fraction** must be vaccinated to prevent it?

All from one or two simple parameters about the disease.

## What is computational epidemiology?

- **Definition.** An interdisciplinary field that combines mathematics, computer science, data science, and epidemiology to model, analyse, and predict health-related dynamics in populations.
- **Core components.**
  - Mathematical modelling of disease-spread dynamics
  - Computer simulations of population-level interactions
  - Data analysis techniques for outbreak detection and tracking
  - Predictive algorithms for forecasting epidemic trajectories
- **Goal.** Develop quantitative tools and simulations that support **evidence-based decision-making** in public health — across **infectious diseases, chronic conditions, environmental exposures, and behavioural trends**.

## Why it matters

- **Informed decision-making** — evidence-based guidance for public-health officials and policymakers during outbreaks (COVID-19 being the textbook example)
- **Intervention design** — test control measures *before* implementing them:
  - Vaccination strategies
  - Social-distancing policies
  - Travel restrictions
  - Resource allocation
- **Preparedness planning** — help healthcare systems anticipate future outbreaks via:
  - Scenario modelling
  - Early-warning systems
  - Resource-requirement forecasts

## Beyond health — same maths, different domains

The same compartmental machinery describes any process where "states" spread through a population:

- **Information diffusion**

- Spread of news, rumours, and misinformation on social media
- Viral transmission of digital content (memes, trends)
- Opinion formation and influence in online communities

- **Cybersecurity**

- Propagation of malware, ransomware, and viruses across networks
- Vulnerability assessment to predict and mitigate risks
- Containment strategies for digital infections

The mathematics learned here transfers cleanly to all of these.

## Two big modelling choices

### Deterministic vs. stochastic models

- **Deterministic** — fixed outcome from initial conditions
  - Built on differential equations (SI, SIS, SIR, SEIR, ...)
  - Good for large populations and analytical insights
- **Stochastic** — outcomes are probability distributions
  - Essential for small populations and the early stages of outbreaks
  - Captures uncertainty but is more computationally expensive

This lecture focuses on **deterministic, ODE-based** compartmental models — the canonical starting point.



## How do contacts shape spread? — Mixing patterns

- **Homogeneous mixing** (mass action)

- Every individual is equally likely to contact every other individual
- Mathematically tractable but **unrealistic** in real populations
- Enables analytical solutions, but oversimplifies actual contact patterns
- **Key blind spot.** Ignores clustering, community structure, and super-spreaders

- **Heterogeneous mixing**

- Individuals have different contact rates and patterns
- Much closer to real-world social interaction networks
- **Critical for real epidemics**, but harder to analyse mathematically
- Requires **network-based** or agent-based approaches
- Often calls for spatially-aware models (geography-based interactions)

This lecture lives in the homogeneous-mixing world. The next lecture (NS23) lifts the assumption.

## Compartmental models — the core idea

- **Definition.** Mathematical models that split the population into distinct **compartments** based on disease status — and track flows between them.
- **Historical roots.**
  - First formalised by Kermack & McKendrick (1927)
  - Revolutionised the understanding of epidemic thresholds
  - Foundation for modern mathematical epidemiology
- **Basic principle.**
  - Individuals move between compartments according to defined rates
  - Transitions are modelled as flows, typically using differential equations
  - Each individual belongs to **exactly one** compartment at a time

## Building blocks

- **States (compartments)**
  - Mutually exclusive categories defined by disease status
  - Common examples: **S**usceptible, **I**nfectious, **R**ecovered
  - Extended versions add: **E**xposed, **V**accinated, ...
- **Transition rules**
  - Mathematical expressions governing flow between compartments
  - Based on biological parameters (infection rate, recovery rate)
  - Typically expressed as rates per unit time

## Building blocks — key parameters

- **Transmission rate**  $\beta$  — rate of infection per contact, per unit time
- **Recovery rate**  $\gamma$  — rate at which infectious individuals leave the infectious state
- **Basic reproduction number**  $R_0 = \beta/\gamma$  — average number of secondary infections caused by one infectious individual in a fully susceptible population

We will return to  $R_0$  once we have a model where it matters (SIS, SIR). For now, just remember:  $R_0 > 1 \rightarrow$  **outbreak**;  $R_0 < 1 \rightarrow$  **fizzles out**.

## Mathematical formulation.

- Typically systems of **ordinary differential equations (ODEs)**
- Sometimes stochastic versions for small populations

## Standard assumptions — what the maths quietly buys

- **Homogeneous mixing** — all individuals interact randomly with equal probability; no spatial structure or clustering of contacts
- **Markov property** — the future state depends only on the current state, not on past history; transition probabilities are constant over time
- **Population-level focus** — large-population approximation; individual variations are averaged out; deterministic models work with expected values
- **No spatial component** — geography and movement patterns generally ignored

Each of these will be relaxed by some extension in Part 5 — and all of them by network models in NS23.

## A roadmap through the models

We will build complexity one compartment at a time. Each model answers a different question:

| Model | Adds                        | Question it answers                                  |
|-------|-----------------------------|------------------------------------------------------|
| SI    | nothing — only S and I      | How fast can a one-shot, no-recovery process spread? |
| SIS   | recovery without immunity   | Does the disease persist (endemic) or die out?       |
| SIR   | recovery with full immunity | When does the epidemic peak? How big does it get?    |
| SEIR  | latent / exposed period     | What if there is a delay before infectiousness?      |
| SIRS  | waning immunity             | What if immunity is temporary?                       |

Each step is a single new compartment or transition. The  $R_0$  vocabulary stays the same throughout.

## Meet the simplest possible epidemic — the SI model

We start with the **simplest** model that captures the essential shape of an outbreak. Just two compartments, one parameter.

**The story.** People are either healthy (S) or sick (I). When an S meets an I, there is some chance the S gets infected. **Once infected, you stay infected forever** (no recovery).

This obviously does not describe any real disease — but it captures the dynamics of any *one-shot* spreading process: chronic infections, computer viruses, irreversible information cascades.

The mathematical machinery developed here will carry over almost unchanged to the more realistic models that follow.

## The SI model

- **S** — susceptible
- **I** — infectious
- **Total population**  $N = S + I$
- **Transmission rate**  $\beta$  (per contact, per unit time)

**Once infected, always infected** — no recovery compartment.

**Differential equations:**

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N}, \\ \frac{dI}{dt} &= \beta \frac{SI}{N}.\end{aligned}$$

The rate of new infections is proportional to the probability a random contact is between an S and an I — hence the  $SI/N$  product.



## Why differential equations for epidemics?

### Mathematical justification

- Epidemics are **dynamic processes** that evolve continuously over time
- Changes in compartment sizes are **rate-based processes**
- For large populations, discrete changes look **continuous**
- ODEs efficiently capture the **deterministic average behaviour** at scale

### Practical benefits

- Well-established mathematical theory (stability, equilibria)
- Efficient numerical methods for solution
- Clear relationship to underlying biological parameters
- Closed-form analytical insights (thresholds, final size)

## SI — analytical solution

The SI equations have an exact closed form:

$$I(t) = \frac{I_0 N e^{\beta t}}{N - I_0 + I_0 e^{\beta t}},$$

where  $I_0 = I(0)$  is the initial number of infectious individuals. This is the classic **logistic curve** — slow start, exponential middle, saturating end.

(It can be derived by separation of variables; the algebra is standard but skipped here.)

**Susceptible side**, by conservation:  $S(t) = N - I(t)$ .

Since there is no recovery, the entire population eventually becomes infected as  $t \rightarrow \infty$ .

## SI — numerical approximation (Euler's method)

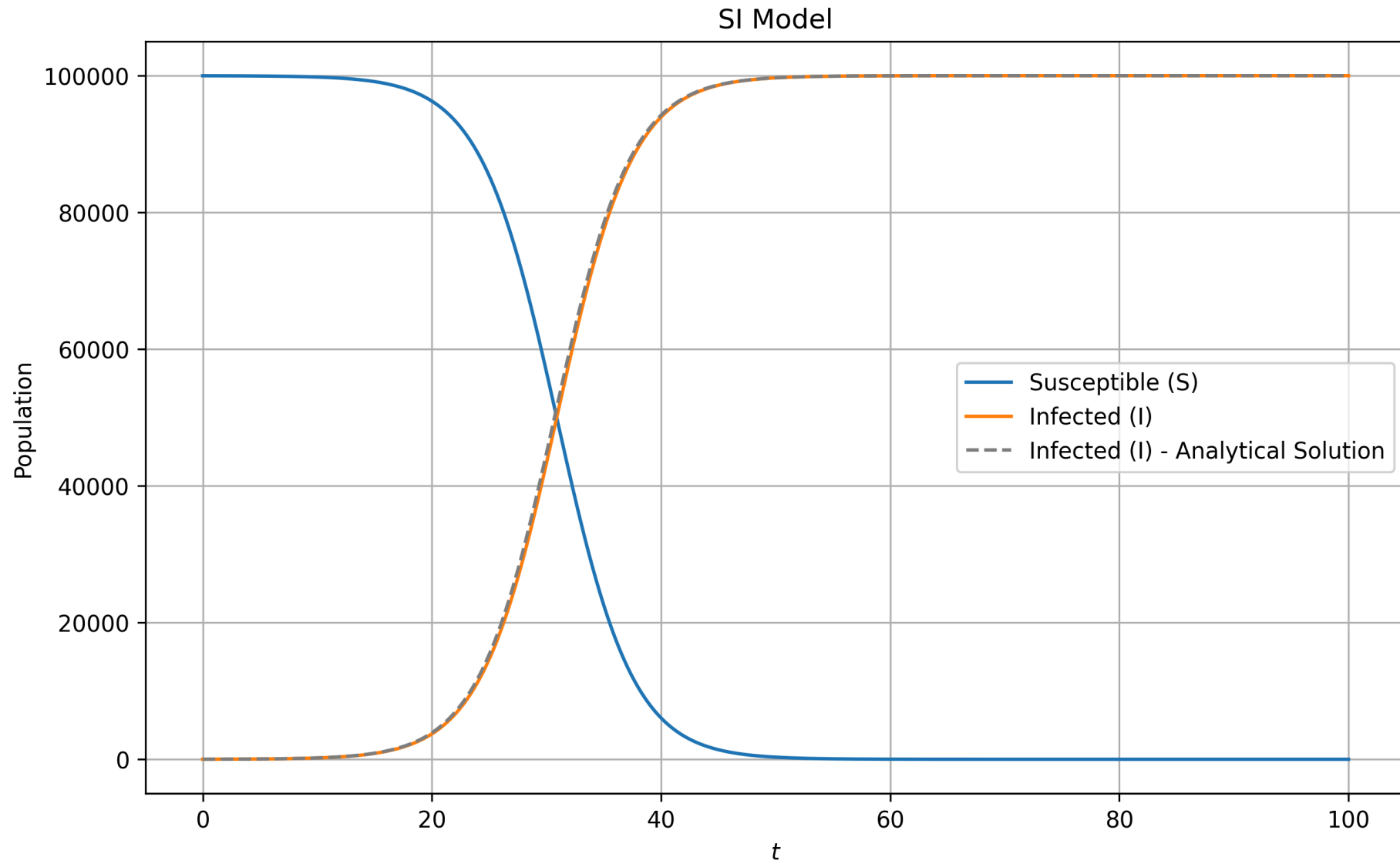
When an exact solution is awkward, Euler's method gives a simple discrete update:

$$\begin{aligned}S(t + \Delta t) &= S(t) - \Delta t \cdot \beta \frac{S(t)I(t)}{N}, \\I(t + \Delta t) &= I(t) + \Delta t \cdot \beta \frac{S(t)I(t)}{N}.\end{aligned}$$

### Recipe:

1. Choose a small time step  $\Delta t$
2. Initialise  $S(0) = N - I_0$  and  $I(0) = I_0$
3. Iteratively compute  $S(t)$  and  $I(t)$  for each time step

This is the workhorse method behind nearly every interactive epidemic simulator.



## Early-stage exponential growth

At the very start of an epidemic, almost everyone is susceptible:  $S \approx N$ . The equation for  $I$  simplifies:

$$\frac{dI}{dt} = \beta \frac{SI}{N} \approx \beta \frac{NI}{N} = \beta I.$$

This is a linear ODE with solution

$$I(t) = I_0 e^{\beta t}.$$

What this means:

- **Exponential growth** — new infections are proportional to current infections
- **Only valid early** — once  $S$  depletes, growth slows (logistic shape)
- **Implication** — early intervention has outsized impact, because it acts during the exponential phase

## SI — key time scales

Two simple metrics summarise the speed of an SI epidemic:

- **Doubling time**  $t_d$  — time for the number of infectious individuals to double during the early exponential phase

$$t_d = \frac{\ln 2}{\beta}$$

- *Smaller* doubling time  $\rightarrow$  faster epidemic
- Used to assess the urgency of interventions
- *Examples:*  $\beta = 0.1/\text{day} \Rightarrow t_d \approx 7 \text{ days}$ ;  $\beta = 0.5/\text{day} \Rightarrow t_d \approx 1.4 \text{ days}$ ;  $\beta = 1.0/\text{day} \Rightarrow t_d \approx 0.7 \text{ days}$

- **Half-infection time**  $t_{50\%}$  — time for half the population to become infected

$$t_{50\%} = \frac{1}{\beta} \ln \left( \frac{N - I_0}{I_0} \right)$$

- Indicates how fast the epidemic reaches critical mass
- Useful for forecasting healthcare demand

Both are most informative early in the epidemic, when interventions can still bend the trajectory.

## SI — applications and takeaway

### Applications

- **Permanent infections** — diseases or conditions where recovery is not possible (some chronic infections)
- **Digital contagion** — computer viruses, irreversible information cascades
- **Baseline model** — the foundation for understanding more complex epidemic dynamics

### Key takeaway

The SI model provides the simplest framework for understanding the dynamics of irreversible spreading processes — capturing the essential exponential-then-saturating shape of any contagion process, biological or digital.

## A more realistic story — recovery without immunity

Real diseases let people recover. The simplest way to add this: introduce a **recovery rate**  $\gamma$  that sends infectious people **back to the susceptible pool**.

- *Recover, become healthy, can be infected again*
- Examples: the common cold, gonorrhoea, many bacterial infections

Note what is interesting: the disease can now have **two possible long-term futures** — die out, or settle into a permanent endemic state. Which of the two happens turns out to be controlled by a single number, which we will name in a few slides.



## The SIS model

- **S** — susceptible, **I** — infectious, then **back to S**
- **Total population**  $N = S + I$  (constant)
- **Transmission rate**  $\beta$ , **recovery rate**  $\gamma$

**"Get sick, recover, get sick again."** Individuals recover but do not gain immunity — they can be infected multiple times. (Think the common cold, or many sexually transmitted infections.)

### Differential equations:

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} + \gamma I, \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I.\end{aligned}$$

The new  $+\gamma I$  term moves recovered individuals straight back into S.

## SIS — analytical solution and numerical update

Closed form for the infectious population:

$$I(t) = \frac{K I_0 e^{(\beta-\gamma)t}}{K + I_0(e^{(\beta-\gamma)t} - 1)}, \quad K = N \left(1 - \frac{1}{R_0}\right).$$

The constant  $K$  is the long-run number of infectious individuals when  $R_0 > 1$  — the **endemic equilibrium**.

**Euler step** for simulation:

$$\begin{aligned} S(t + \Delta t) &= S(t) - \Delta t \cdot \left( \beta \frac{S(t)I(t)}{N} - \gamma I(t) \right), \\ I(t + \Delta t) &= I(t) + \Delta t \cdot \left( \beta \frac{S(t)I(t)}{N} - \gamma I(t) \right). \end{aligned}$$

## The hero of this lecture — meet $R_0$

Before any equations, ask the question that drives everything that follows:

*On average, how many other people does one infectious person infect, in a fully susceptible population?*

That single number — the **basic reproduction number**  $R_0$  — is the most important quantity in epidemiology. It decides:

- whether the disease will spread or die out
- how big the epidemic gets
- when the peak hits
- how much vaccination is needed

Almost every public-health intervention you have ever heard of (masks, distancing, vaccination, quarantine) can be understood as **trying to push  $R_0$  below 1**.

## The basic reproduction number $R_0$ — the central concept

**Definition.**  $R_0$  is the average number of secondary infections caused by a single infectious individual in a fully susceptible population. For homogeneous-mixing models:

$$R_0 = \frac{\beta}{\gamma}$$

- $\beta$  — transmission rate per contact, per unit time
- $\gamma$  — recovery rate (so  $1/\gamma$  is the average infectious period)

### Why it matters — the epidemic threshold.

- $R_0 < 1$  — each infected person produces *fewer* than one new infection on average → the disease dies out
- $R_0 > 1$  — each infected person produces *more* than one new infection → the disease can grow exponentially → outbreak
- $R_0 = 1$  is the **epidemic threshold**, separating the two regimes

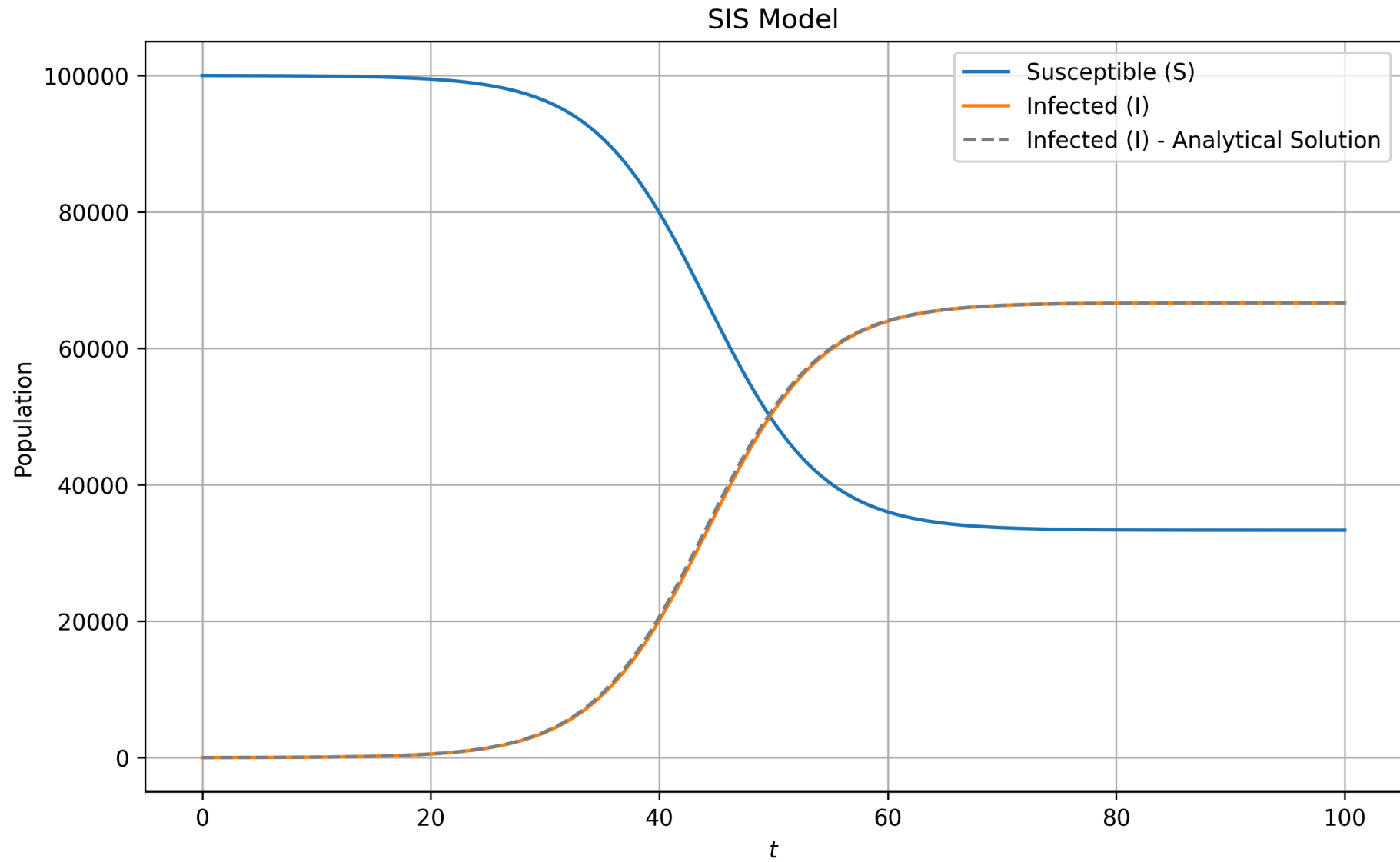
This single number is the most important quantity in mathematical epidemiology — it appears in every model from here on.

## $R_0$ in practice — some real-world numbers

$R_0$  is used to estimate the intensity of interventions needed to control an epidemic (vaccination coverage, social distancing) and to compare the spread potential of different pathogens.

| Disease                    | Approximate $R_0$ |
|----------------------------|-------------------|
| Measles                    | 12 – 18           |
| Pertussis (whooping cough) | 5 – 17            |
| Smallpox                   | 5 – 7             |
| SARS-CoV-2 (ancestral)     | 2 – 3             |
| Ebola                      | 1.5 – 2.5         |
| Seasonal influenza         | 1.2 – 1.8         |

**Key takeaway.** Estimating  $R_0$  is the first step in designing any public-health response — it determines what fraction of the population needs to be immune to stop transmission.



## Threshold behaviour in SIS

The reproduction number  $R_0 = \beta/\gamma$  controls the long-run fate of the disease.

If  $R_0 < 1$

- The disease cannot sustain itself
- Infected individuals recover faster than they can infect others
- The population converges to the **disease-free equilibrium** ( $I^* = 0$ )

If  $R_0 > 1$

- The disease spreads and persists in the population
- The population converges to the **endemic equilibrium** with

$$\frac{I^*}{N} = 1 - \frac{1}{R_0}$$

This is the first place where  $R_0$  does real work — it doesn't just decide *whether* the disease persists, it sets the *level* at which it persists.

## SIS — key metrics

- **Growth rate.** During the early phase, with  $S \approx N$ :

$$\frac{dI}{dt} \approx (\beta - \gamma)I = \gamma(R_0 - 1)I$$

The coefficient  $r = \gamma(R_0 - 1)$  controls how fast infections grow (or decline if  $R_0 < 1$ ).

- **Time to equilibrium.** Approximate time to settle:

$$t_{\text{eq}} \approx \frac{5}{|r|} = \frac{5}{|\beta - \gamma|}$$

- **Endemic fraction infected** (for  $R_0 > 1$ ):

$$\frac{I^*}{N} = 1 - \frac{1}{R_0}$$

A higher  $R_0$  pushes more of the population into the chronically infected pool.

A few values to build intuition:

| $R_0$ | Endemic infected fraction $I^*/N$ |
|-------|-----------------------------------|
| 1.5   | 33%                               |



## Adding lasting immunity — the SIR story

Now we make recovery **permanent**: once you have had the disease, you cannot catch it again. We add a third compartment for that:

- **R — Recovered.** Has had the disease and is now immune (or removed — death, isolation).

This is the model of choice for almost every "wave-shaped" epidemic you have seen on the news — measles, COVID-19, most childhood diseases.

**Why this small change is huge.** In SIS the susceptible pool is constantly refilled, so the disease can persist forever. In SIR, **every recovery permanently shrinks the susceptible pool**. Eventually  $S$  becomes too small to sustain transmission, and the epidemic ends — *automatically*, without any intervention.

## The SIR model

- **S** — susceptible, **I** — infectious, **R** — recovered/removed
- **Total population**  $N = S + I + R$
- **Transmission rate**  $\beta$ , **recovery rate**  $\gamma$
- Recovery confers **permanent immunity** (or removal — death, isolation)

### Differential equations:

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N}, \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I, \\ \frac{dR}{dt} &= \gamma I.\end{aligned}$$

The third compartment makes this the canonical model for epidemics with lasting immunity (most childhood diseases, COVID-19 in approximation).

## SIR — no closed form, so we simulate

Unlike SI and SIS, the SIR model has **no closed-form solution** because the equations are coupled non-linearly. We solve them numerically with the Euler step:

$$\begin{aligned}S(t + \Delta t) &= S(t) - \Delta t \cdot \beta \frac{S(t)I(t)}{N} \\I(t + \Delta t) &= I(t) + \Delta t \cdot \left( \beta \frac{S(t)I(t)}{N} - \gamma I(t) \right) \\R(t + \Delta t) &= R(t) + \Delta t \cdot \gamma I(t)\end{aligned}$$

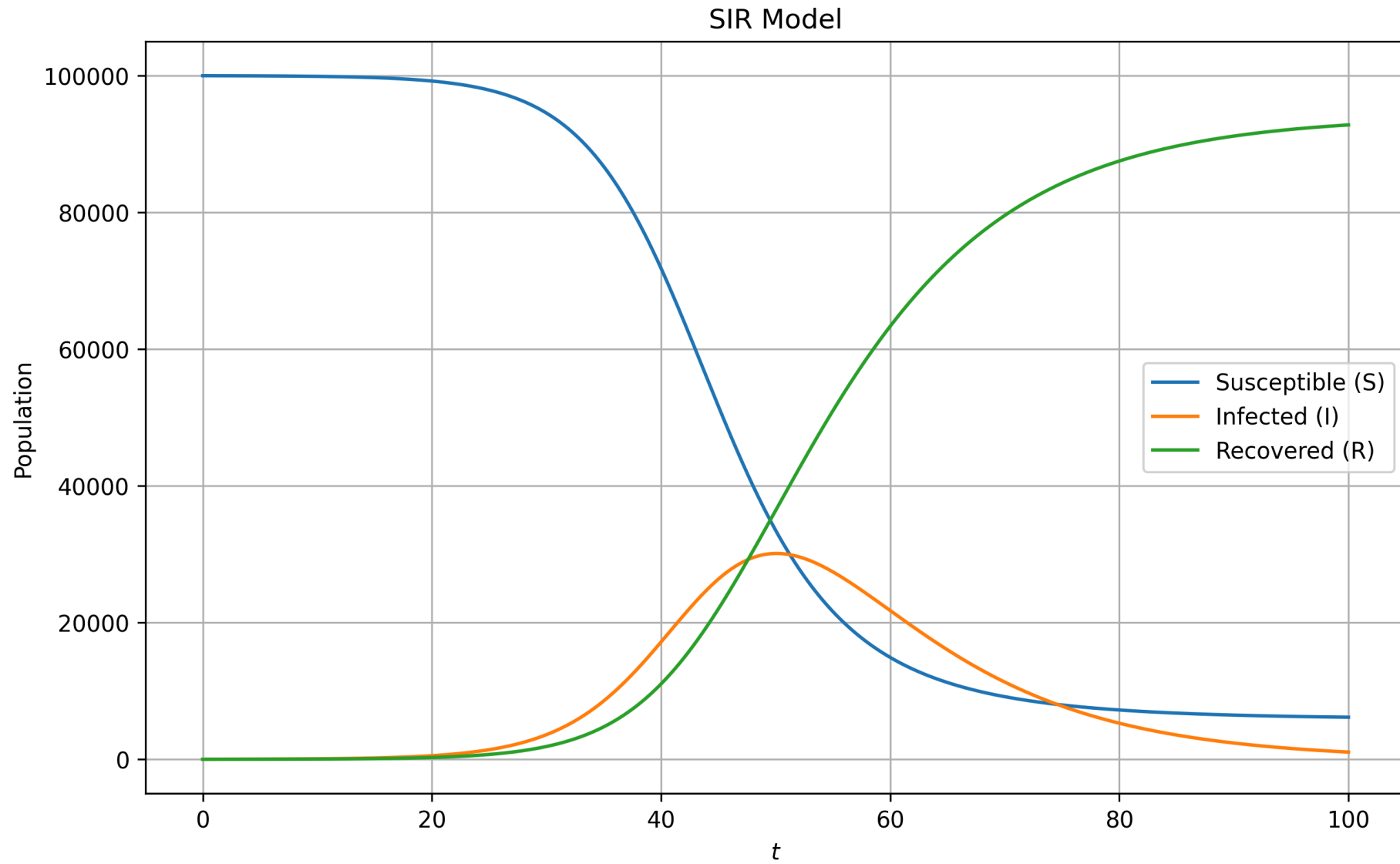
Despite the lack of a closed form, several **analytical insights** can be extracted directly from the equations — they are the subject of the next few slides.

## $R_0$ in SIS vs. SIR — same formula, different consequence

Both models share  $R_0 = \beta/\gamma$ . The interpretation differs:

- **SIS** —  $R_0$  controls disease *persistence*
  - $R_0 < 1$ : dies out
  - $R_0 > 1$ : becomes **endemic** (settles at  $I^*/N = 1 - 1/R_0$ )
- **SIR** —  $R_0$  controls *whether an epidemic occurs at all*
  - $R_0 < 1$ : no epidemic
  - $R_0 > 1$ : epidemic occurs, peaks, then ends as susceptibles run out

The SIR epidemic is **self-limiting**: even without intervention, it eventually ends because the susceptible pool empties.



## Three phases of an SIR epidemic

The dynamics naturally split into three phases, governed by where  $S$  sits relative to the critical level  $S_c = N/R_0$ .

### Growth phase — $S > N/R_0$

- $\frac{dI}{dt} > 0$ : infections exceed recoveries
- **Initial exponential growth** in the infectious population

### Peak phase — $S = N/R_0$

- $\frac{dI}{dt} = 0$ : infections balance recoveries
- **Maximum number of simultaneous infections** is reached

### Decline phase — $S < N/R_0$

- $\frac{dI}{dt} < 0$ : recoveries exceed infections
- **Epidemic naturally subsides**

The critical susceptible level  $S_c = N/R_0$  is the same threshold that drives every SIR result that follows.

## Key measure — attack rate

**Attack rate** = the final fraction of the population that gets infected during the epidemic:

$$\text{Attack rate} = 1 - \frac{S_{\infty}}{N}$$

- $S_{\infty}$  = number of susceptible individuals remaining at the end of the epidemic — those who were never infected (naturally resistant, escaped exposure, or vaccinated)
- **Interpretation.** Total impact of the epidemic — how many people were ultimately infected
- **Implication.** Higher  $R_0 \rightarrow$  higher attack rate  $\rightarrow$  stronger case for early intervention

## Final size relation — predicting the total without simulating

The **final size relation** links the epidemic's eventual size to  $R_0$  alone, without solving the full ODE system:

$$\ln\left(\frac{S_0}{S_\infty}\right) = R_0\left(1 - \frac{S_\infty}{N}\right)$$

This is an **implicit** equation in  $S_\infty$  — solve it numerically, and you know the final attack rate.



## What it tells us, qualitatively:

- Larger  $R_0 \rightarrow$  smaller  $S_\infty \rightarrow$  larger attack rate
- Even at very large  $R_0$ ,  $S_\infty$  does not reach zero — some fraction always escapes infection
- A tool for long-term planning and for evaluating interventions: any policy that effectively reduces  $R_0$  shrinks the final epidemic size predictably

A few worked values:

| $R_0$ | Attack rate $\approx$ |
|-------|-----------------------|
| 1.5   | 58%                   |
| 2     | 80%                   |
| 3     | 94%                   |
| 5     | 99%                   |

## Key measure — peak prevalence

**Peak prevalence** = the maximum fraction of the population infected *simultaneously*. A useful approximation:

$$\text{Peak prevalence} \approx 1 - \frac{1 + \ln R_0}{R_0}$$

- **Interpretation.** Worst-case burden on healthcare systems — the highest number of active cases at any single moment
- **Implication.** "**Flattening the curve**" reduces peak prevalence, even if total cases (attack rate) end up similar — buying time for hospital capacity, vaccines, treatments

The same total number of cases spread over twice the time can be the difference between a manageable response and a collapsing health system.

A few worked values:

| $R_0$ | Peak prevalence |
|-------|-----------------|
| 1.5   | ~5%             |
| 2     | ~15%            |

## "Flatten the curve" in one slide

Same disease, two scenarios:

- **No intervention** — epidemic ends fast, but the peak is huge and overwhelms hospitals
- **With intervention** (lower  $\beta$  via masks, distancing, etc.) — epidemic lasts longer, but the peak is much lower

Total cases (attack rate) may end up similar — but the **distribution over time** is what determines whether the health system survives.

This is what the entire 2020 "flatten the curve" public-health campaign was about.

## Key measure — doubling time

**Doubling time** = how long it takes for case counts to double during the early exponential growth phase:

$$t_d \approx \frac{\ln 2}{\gamma(R_0 - 1)}$$

- **Interpretation.** A shorter doubling time means a faster-growing epidemic
- **Implication.** Critical for assessing the urgency of interventions during the early phase, when news headlines and case counts move fastest

This is the number that drove most early COVID-19 reporting in 2020.

## Early growth approximation

When  $S \approx S_0$  and  $I$  is small, infections grow roughly exponentially:

$$I(t) \approx I_0 e^{(\beta \frac{S_0}{N} - \gamma) t} = I_0 e^{\gamma(R_0 - 1) t}$$

- **Interpretation.** Same logistic shape as SI early on — the recovery term just shifts the exponent
- **Implication.** Early intervention slows the exponent, with compounding effects over time

## Peak infection time

The infection rate **peaks** when the rate of new infections balances the rate of recoveries:

$$\frac{dI}{dt} = 0 \implies S = \frac{\gamma N}{\beta} = \frac{N}{R_0}$$

- **Interpretation.** The peak occurs the moment  $S$  falls below  $N/R_0$  — the critical threshold introduced earlier
- **Implication.** Knowing *when* the peak hits helps plan healthcare capacity and intervention timing

## Control parameter — herd immunity

**Herd immunity** occurs when a sufficient fraction of the population is immune that the disease cannot spread to the remaining susceptibles.

- **Herd immunity threshold:**

$$p_c = 1 - \frac{1}{R_0}$$

The minimum *immune fraction* needed to prevent epidemic spread. Higher  $R_0$  requires higher immunity.

- **Vaccination coverage needed** (in a fully susceptible starting population):

$$v_c = \frac{R_0 - 1}{R_0}$$

- *Worked example.* If  $R_0 = 4$ , then  $v_c = 3/4 = 75\%$  — three out of four people must be vaccinated to prevent outbreaks.

A few values to feel the formula:

| $R_0$        | Required immune fraction $p_c$ |
|--------------|--------------------------------|
| 2            | 50%                            |
| 4            | 75%                            |
| 10           | 90%                            |
| 15 (measles) | ~93%                           |

This single formula explains why measles ( $R_0 \sim 15$ , requires ~95% coverage) is so much harder to eliminate than influenza — and why the anti-vaccine movement is such a public-health concern: even small drops in coverage can push us back below the threshold.



## Key insights from the SIR model

- **Epidemic threshold.** The disease spreads only when  $R_0 > 1$ . Small changes in transmission near the threshold have outsized effects on outcomes.
- **Self-limiting nature.** Unlike SIS, SIR epidemics naturally end as the susceptible pool depletes. Final size is set by  $R_0$  and initial conditions.
- **Intervention timing.** Early interventions are by far the most effective — by the time the peak is reached, most infections have already happened.
- **Flattening the curve.** Reducing  $\beta$  extends the epidemic's duration while lowering its peak. Total cases may end up similar, but spread over a longer time — protecting healthcare capacity.

## When SIR isn't enough — extending the model

Real diseases have features SIR alone cannot capture:

- An **incubation period** before you become infectious
- **Waning immunity** — protection that fades over time
- **Births and deaths** in long-term endemic diseases
- **Age-dependent** transmission and severity

Each is handled by adding a compartment or a flow. Same machinery, more biology — the next few slides survey the canonical extensions.

## SEIR — adding a latent (Exposed) period

- **Adds an Exposed (E) compartment** for individuals who are infected but **not yet infectious** (in the latent / incubation period)
- **Use case.** Diseases with a significant incubation time (COVID-19, measles)
- **Dynamics.**
  - $S \rightarrow E$ : susceptibles become exposed upon infection
  - $E \rightarrow I$ : exposed become infectious after the incubation period
  - $I \rightarrow R$ : infectious recover
- **Practical insights.**
  - Captures delays in disease spread
  - Crucial for modelling **quarantine** during the latent phase, when a person is infected but cannot yet infect others

## SIRS — adding waning immunity

- **Adds an  $R \rightarrow S$  flow** — recovered individuals lose immunity over time and return to the susceptible pool
- **Use case.** Diseases where immunity is temporary (seasonal flu, common cold, many reinfectable diseases)
- **Dynamics.** Same as SIR plus waning rate
- **Practical insights.**
  - Models recurring outbreaks and seasonal epidemics
  - Highlights the importance of **booster vaccinations**

## SEIRS — combining latency and waning immunity

- **Combines SEIR and SIRS** — both a latent period *and* waning immunity
- **Use case.** Diseases with incubation periods and temporary immunity (COVID-19 with waning immunity is a well-studied recent example)
- **Practical insights.**
  - Captures complex dynamics of reinfection and delayed spread
  - Used for long-term epidemic forecasting

## SIR with demography — births and deaths

- **Adds population turnover** — births replenish the susceptible pool, natural deaths remove individuals from all compartments
- **Use case.** Long-term modelling of endemic diseases (measles, malaria, pertussis)
- **Dynamics.** Births inflow into  $S$ ; natural deaths drain  $S$ ,  $I$ , and  $R$  at the same per-capita rate
- **Practical insights.**
  - Models the **steady-state** behaviour of diseases in populations with constant turnover
  - Highlights the role of **routine vaccination** in maintaining herd immunity over generations

## Age-structured SIR

- **Adds age-specific compartments** —  $S$ ,  $I$ , and  $R$  are split into age groups, each with its own contact rate and disease dynamics
- **Use case.** Diseases with strong age-dependent transmission or severity (COVID-19, influenza)
- **Practical insights.**
  - Captures heterogeneity in spread and outcomes across age groups
  - Enables **targeted interventions** — e.g. vaccinating high-risk age groups first

This extension is one step toward heterogeneity, but it still treats all members within an age group as exchangeable.

## The mean-field assumption — and why it eventually breaks

Every model in this lecture has assumed **homogeneous mixing**: anyone can infect anyone with equal probability. That assumption buys mathematical tractability — but at a real cost.

### What homogeneous mixing misses:

- **Network structure** — friendships, families, workplaces, schools shape *who* you contact
- **Hubs and super-spreaders** — a few highly connected people drive disproportionate transmission
- **Clustering** — your contacts know each other, so infections "double back" within communities
- **Spatial / geographic structure** — outbreaks travel along physical contact patterns

A high-school superspreader and a hermit are mathematically identical in mean-field. In reality, they are not.



## Preview — what changes on a network (NS23)

Once we replace homogeneous mixing with an explicit contact network, several quantities change in important ways:

- The epidemic threshold becomes  $\beta_c = \frac{\langle k \rangle}{\langle k^2 \rangle} \cdot \gamma$  rather than  $\beta_c = \gamma / \langle k \rangle$
- For **scale-free networks** (with broad, heavy-tailed degree distributions),  $\langle k^2 \rangle$  can grow without bound — making  $\beta_c \rightarrow 0$
- **Translation.** On a real social network, the epidemic threshold can effectively *vanish* — even very weakly transmissible diseases can spread
- **Intervention.** Targeting **hubs** (high-degree individuals) for vaccination becomes much more efficient than random vaccination

These results — and the friendship paradox that lets us *find* hubs without knowing the network — are the subject of NS23.

Q & A

